

Optimization of thermal transport properties in multilayer silicon structures via a hybrid photoacoustic and PINN approach

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Understanding heat transport in multilayer porous silicon (MPSi) structures is of great importance for modern micro- and nanoelectronics, thermoelectric systems, and advanced thermal insulation technologies, where precise control of thermal conductivity is essential to ensure device reliability. While single-layer porous silicon offers significant thermal-conductivity reduction, MPSi structures introduce complex phonon-scattering mechanisms at interfaces between layers of varying porosity. To address these challenges, this study reveals a comprehensive multiscale approach combining experimental measurements, molecular dynamics (MD) simulations, and machine learning.

The thermal transport properties of electrochemically etched MPSi structures with alternating porosity layers (ranging from 50% to 75%) were investigated using the gas-microphone photoacoustic (PA) method. The non-destructive PA technique provided the amplitude-frequency characteristics of signals, which were analyzed to determine the effective thermal conductivity of the structures.

To overcome the limitations of classical effective medium approximations, which fail to fully capture volumetric light absorption and morphological imperfections, a Physics-Informed Neural Network (PINN) was employed. The PINN models were trained by integrating the 2D Fourier heat conduction equation directly into the loss function, using realistic structural inputs generated via the Quartet Structure Generation Set algorithm, and were validated by Scanning Electron Microscopy. Furthermore, MD simulations utilizing the Green-Kubo method were performed to investigate the microscopic mechanisms of phonon transport. Both experimental and PINN-based data deviate from standard effective medium predictions, highlighting the critical role of interfacial thermal boundaries. This hybrid approach provides effective predictive models for the precise design and optimization of advanced thermal management components based on nanostructured silicon.

The work was supported by National Research Foundation of Ukraine (project No. 2023.03/0252).